Computer Engineering Laws and Professional Ethics

I. Computer Engineering as a Discipline

D. Core Areas of Study

4. Operating Systems

Operating Systems are system software that act as an intermediary between computer hardware and users or application programs. They manage hardware resources and provide essential services that enable efficient and secure system operation.

a. Resource Management

The OS is responsible for allocating and controlling system resources to ensure optimal performance:

- **CPU management**: Determines which process uses the processor and for how long.
- **Memory management**: Allocates RAM to programs, manages virtual memory, and prevents memory conflicts.
- **Storage management**: Organizes files, controls disk access, and manages secondary storage devices.

  Efficient resource management maximizes system utilization while minimizing delays and conflicts.

b. Process Scheduling and Multitasking

The OS supports **multitasking**, allowing multiple programs to run seemingly at the same time:

- Uses **scheduling algorithms** (e.g., First-Come-First-Serve, Round Robin, Priority Scheduling).
- Manages **process states** (ready, running, waiting, terminated).
- Enables **context switching** to quickly switch the CPU between processes.

  This ensures responsiveness and fairness among users and applications.

c. User-System Interface

The OS provides interfaces through which users interact with the computer:

- **Command-Line Interface (CLI)** for text-based commands.
  - Characteristics
  - Advantages
  - Limitations
- **Graphical User Interface (GUI)** for visual interaction using icons and menus.
  - Characteristics
  - Advantages
  - Limitations
- Importance of a Good User Interface.
  - Improved Usability
  - Enhanced Accessibility
  - Better System Control

5. Computer Networks

Computer Networks involve the interconnection of computers and devices to share data, resources, and services efficiently and securely.

A **computer network** is a collection of interconnected devices (computers, servers, routers, smartphones, IoT devices) that exchange data and share resources.

Networks enable:

- Internet access
- File sharing
- Communication (email, video calls)
- Cloud services
- Smart systems (IoT, automation)

a. Data transmission and protocols.

- Data Transmission

  Data transmission refers to the process of sending digital information from one device to another across a network.

  - Key Elements of Data Transmission
    - ✓ Sender – device that transmits data
    - ✓ Receiver – device that receives data
    - ✓ Medium – path used for transmission
    - ✓ Wired: Ethernet cables, fiber optics
    - ✓ Wireless: radio waves, satellites
    - ✓ Data Signals – electrical or electromagnetic signals
    - ✓ Protocols – rules that control communication

- Packet Switching

  Most networks transmit data using packet switching:

  - Large data is broken into small packets
  - Each packet contains:
    - ✓ Source address
    - ✓ Destination address
    - ✓ Data payload
    - ✓ Error-checking bits
  - Packets travel independently and are reassembled at the destination

  This makes communication faster and efficient.

- Protocols in Networking

  A protocol is a set of rules that ensures devices communicate correctly.

  Protocols define:

  - Data format
  - Transmission speed
  - Error handling
  - Acknowledgment
  - Routing

- Important Protocols

  TCP/IP (Transmission Control Protocol / Internet Protocol). TCP/IP is the foundation of the Internet.

  - IP (Internet Protocol)
    - ✓ Handles addressing and routing
    - ✓ Ensures packets reach the correct destination
    - ✓ Uses IP addresses (e.g., 192.168.1.1)
  - TCP (Transmission Control Protocol)
    - ✓ Provides reliable communication
    - ✓ Establishes a connection before sending data
    - ✓ Ensures packets arrive in correct order
    - ✓ Retransmits lost packets
  - Used for:
    - ✓ Web browsing
    - ✓ Email
    - ✓ File transfers

- HTTP and HTTPS

  HTTP (Hypertext Transfer Protocol)

- Used for accessing websites
- Transfers web pages between browser and server
- Not encrypted

Example: http://example.com

- HTTPS (Secure HTTP)
  - Encrypted version of HTTP
  - Uses SSL/TLS encryption
  - Protects sensitive information

  Used for:
  - Online banking
  - E-commerce
  - Secure logins

  Example: https://example.com
- UDP (User Datagram Protocol)

  UDP is a connectionless protocol.
  - No connection setup
  - Faster but less reliable
  - No acknowledgment or retransmission

  Used for:
  - Video streaming
  - Online gaming
  - Voice calls (VoIP)

b. Network security and management.

Networks must be protected from unauthorized access and attacks. Networks are vulnerable to threats, so protecting them is essential.

- Network Security

  Network security refers to protecting network systems and data from unauthorized access, misuse, or attacks.
  - Common Network Threats
    - ✓ Hacking – unauthorized access
    - ✓ Malware – viruses, worms, ransomware
    - ✓ Phishing – stealing credentials
    - ✓ DoS/DDoS attacks – flooding network traffic
    - ✓ Man-in-the-middle attacks – intercepting communication
  - Security Mechanisms

    Firewalls

    A firewall acts as a security gate.
    - ✓ Filters incoming and outgoing traffic
    - ✓ Blocks unauthorized access
    - ✓ Can be hardware or software-based
  - Encryption

    Encryption converts data into unreadable form.
    - ✓ Only authorized users can decrypt it
    - ✓ Ensures privacy and confidentiality

    Example: HTTPS uses encryption.
  - Authentication

    Authentication verifies identity.

    Examples:
    - ✓ Passwords
    - ✓ Biometrics
    - ✓ Multi-factor authentication (MFA)
  - Intrusion Detection Systems (IDS)

    IDS monitors network activity for suspicious behavior.
    - ✓ Detects attacks early
    - ✓ Alerts administrators

- Access Control

  Controls who can use the network.

  Examples:

  ✓ User permissions
  ✓ Role-based access
  ✓ Network segmentation

- CIA Triad in Network Security

  Security focuses on:

  ✓ Confidentiality – only authorized access
  ✓ Integrity – data is not altered
  ✓ Availability – network services remain accessible

- Network Management

  Network management ensures smooth and efficient network operation.

  - Main Network Management Tasks

    ✓ Monitoring traffic
    ✓ Managing network devices (routers, switches)
    ✓ Optimizing performance
    ✓ Troubleshooting faults
    ✓ Updating firmware and configurations

  - Network Monitoring Tools

    Engineers use tools like:

    ✓ SNMP (Simple Network Management Protocol)
    ✓ Wireshark (packet analysis)
    ✓ Network performance dashboards

c. *Internet and wireless communication.*

Modern networks include both wired and wireless technologies:

- **Internet technologies**
  - **How the Internet Works**

    ✓ Devices connect through ISPs
    ✓ Data travels through routers and backbone networks
    ✓ Uses TCP/IP for communication

  - **Key Internet Services**

    ✓ Web browsing
    ✓ Cloud computing
    ✓ Email
    ✓ Social media
    ✓ Remote work systems

  - **Cloud Computing and Networks**

    Networks enable cloud services such as:

    ✓ Google Drive
    ✓ AWS
    ✓ Microsoft Azure
    ✓ Data is stored and accessed remotely through the Internet.

- **Wireless Communication**

  Wireless networks transmit data without cables using radio waves.

  - Major Wireless Technologies

    Wi-Fi

    ✓ Wireless LAN technology
    ✓ Used in homes, schools, offices
    ✓ Based on IEEE 802.11 standards

    Bluetooth

    ✓ Short-range communication
    ✓ Used for headphones, keyboards, IoT devices
    ✓ Low power consumption

    Cellular Networks (3G, 4G, 5G)

Provide mobile connectivity over wide areas.

- ✓ 4G supports high-speed internet
- ✓ 5G enables:
  - ➢ Ultra-fast speeds
  - ➢ Low latency
  - ➢ Massive IoT connectivity

Satellite Communication

- Used for:
  - ✓ Remote areas
  - ✓ GPS systems
  - ✓ Global broadcasting

Wireless Network Challenges

- Signal interference
- Security risks
- Limited bandwidth
- Range limitations

Engineers solve these using:

- Strong encryption
- Better antennas
- Frequency management
- Advanced protocols

6. Embedded Systems

An embedded system is a special-purpose computer system built inside a larger device to perform a specific task. Unlike a general-purpose computer (like a laptop), an embedded system is designed to do one job very efficiently, such as:

- Controlling a car engine
- Monitoring a patient's heartbeat
- Running a washing machine cycle

These systems are called *embedded* because they are embedded inside a device, not used as standalone computers.

Key Components of Embedded Systems

Embedded systems consist of both hardware and software working together.

a. Microcontroller or Microprocessor

- Microcontroller (MCU) - a single-chip computer that includes:
  - CPU
  - Memory
  - Input/output ports

  It is the most common processor used in embedded systems.

  Example: Arduino boards use microcontrollers.

- Microprocessor (MPU) - a more powerful CPU that usually requires external components like memory and peripherals. Used in complex embedded devices like:
  - A microprocessor is Smartphones
  - Smart TVs
  - Industrial controllers

- Role

  The processor is the "brain" of the embedded system. It:
  - Reads sensor inputs
  - Processes data
  - Sends commands to outputs

b. Memory (RAM, ROM, Flash)

Embedded systems require memory for storing programs and data.

- RAM (Random Access Memory)
  - Temporary storage
  - Used while the system is running
  - Stores variables, sensor readings, and computations

Example: Real-time speed data in a car ECU.

- ROM (Read-Only Memory)

- Permanent storage
- Stores essential boot instructions

- Flash Memory
  - Non-volatile (data stays even when power is off)
  - Stores firmware and updates

  Example: Firmware updates in smart appliances.

c. *Input/Output Interfaces (I/O Interfaces)*

Embedded systems must communicate with the outside world using I/O interfaces.

- Inputs- devices that provide data to the system:
  - Sensors (temperature, pressure, motion)
  - Switches and buttons
  - Cameras and microphones

  Example: A heart monitor sensor sending pulse signals.

- Outputs

  Devices controlled by the system:
  - Motors
  - Displays
  - LEDs
  - Speakers
  - Actuators

  Example: Airbag deployment mechanism.

- Communication Interfaces

  Embedded systems also use protocols like:
  - UART
  - SPI
  - I2C
  - CAN (used in vehicles)

d. *Embedded Software or Firmware*

- Embedded Software - Programs written specifically for the device.
- Firmware - Low-level software stored in ROM or Flash that directly controls hardware. Examples include:
  - BIOS in computers
  - Control logic in washing machines
  - Pacemaker operating software

- Key Characteristics

  Firmware is usually:
  - Highly optimized
  - Reliable
  - Hard to modify once deployed

e. *Constraints of Embedded Systems*

Embedded systems do not have unlimited resources like PCs. They must operate under strict constraints:

- Limited Power

  Many embedded devices run on batteries, so they must consume minimal energy.

  Examples:
  - Wearable fitness trackers
  - Wireless sensors

  Power efficiency is critical.

- Limited Memory and Storage

  Embedded systems often have:
  - Small RAM (kilobytes to megabytes)
  - Limited Flash storage

  So software must be lightweight and efficient.

- Real-Time Performance Requirements

  Many embedded systems must respond instantly.

Real-Time System - A system where correct operation depends on both:
- Correct output
- Correct timing

Example:
- Airbag must deploy within milliseconds during a crash.

- D. Reliability and Safety Requirements

  Embedded systems often run continuously and cannot fail easily.

  Example:
  - Medical infusion pumps must deliver correct doses
  - Aircraft control systems must be fault-tolerant

f. Applications in Devices

Embedded systems are widely used in everyday and critical technologies.

- Automobiles

  Modern vehicles contain dozens of embedded controllers.

  Examples:
  - Engine Control Unit (ECU) - Controls fuel injection, ignition timing, and emissions.
  - ABS (Anti-lock Braking System) - Prevents wheel locking during braking.
  - Airbag Systems - Detect collisions and deploy airbags instantly.

  Importance:

  These are safety-critical embedded systems, meaning failure can cause accidents or loss of life.

- Home Appliances

  Appliances use embedded systems for automation and user convenience.
  - Examples:
    - ✓ Washing machines adjusting water level and cycle timing
    - ✓ Microwave ovens controlling heating time and power
    - ✓ Smart refrigerators monitoring temperature
  - Benefits

    Embedded systems make appliances:
    - ✓ Efficient
    - ✓ Automated
    - ✓ User-friendly

- Medical Equipment

  Healthcare relies heavily on embedded technology.
  - Examples:
    - ✓ Heart monitors tracking vital signs
    - ✓ Infusion pumps delivering controlled medication
    - ✓ Pacemakers regulating heart rhythm
  - Critical Requirement

    Medical embedded systems must be:
    - ✓ Extremely reliable
    - ✓ Accurate
    - ✓ Safe for human life

g. Why Embedded Systems Matter

Embedded systems are essential because they provide:

- Automation in devices
- Real-time control
- Efficient use of hardware
- Reliability in critical applications
- Smart functionality in modern technology

Almost every smart device today depends on embedded computing.

7. Artificial Intelligence

Artificial Intelligence focuses on creating systems that can simulate human intelligence, enabling machines to learn, reason, and make decisions.

a. Machine learning and automation.

- Machine Learning (ML)

  Machine Learning is a major branch of AI that allows systems to learn from data instead of being explicitly programmed. Rather than writing fixed rules, engineers train models using examples so that the system improves performance over time.

  Key Idea

  Machine learning enables computers to learn patterns from experience and make predictions or decisions automatically.

- Types of Machine Learning
  - Supervised Learning

    In supervised learning, the system is trained using labeled data, meaning the correct output is already known.

    ✓ Input data → Output label provided
    ✓ Model learns mapping between them

    Examples:

    ✓ Email spam detection (spam vs not spam)
    ✓ Image classification (cat vs dog)
    ✓ Predicting house prices

    Common algorithms:

    ✓ Linear Regression
    ✓ Decision Trees
    ✓ Support Vector Machines (SVM)
    ✓ Neural Networks
  - Unsupervised Learning

    In unsupervised learning, the system is given unlabeled data and must find patterns or structure on its own.

    ✓ No correct answers provided
    ✓ Model discovers relationships

    Applications:

    ✓ Customer segmentation
    ✓ Data clustering
    ✓ Anomaly detection

    Common techniques:

    ✓ K-Means Clustering
    ✓ Hierarchical Clustering
    ✓ Principal Component Analysis (PCA)
  - Reinforcement Learning

    Reinforcement learning is based on trial and error. An agent learns by interacting with an environment and receiving:

    ✓ Rewards for correct actions
    ✓ Penalties for wrong actions

    The goal is to maximize long-term reward.

    Examples:

    ✓ AI in games (Chess, Go)
    ✓ Robot navigation
    ✓ Self-driving vehicle decision-making

- Pattern Recognition and Predictive Analysis
  - Pattern Recognition

    Pattern recognition refers to AI's ability to detect meaningful patterns in data. It is essential in:

    ✓ Speech recognition
    ✓ Face recognition
    ✓ Handwriting recognition

- ✓ Medical diagnosis

  AI identifies features that humans may not easily detect.

- Predictive Analysis
  Predictive analysis uses historical data to predict future outcomes.

  Examples include:

  - ✓ Weather forecasting
  - ✓ Stock market prediction
  - ✓ Predicting machine failure in factories
  - ✓ Disease outbreak detection

  This is widely used in engineering for optimization and preventive maintenance.

- Automation of Repetitive and Complex Tasks

  Automation means using AI systems to perform tasks with minimal human involvement.

  - Types of tasks automated by AI

    Repetitive tasks:

    - ✓ Data entry
    - ✓ Sorting documents
    - ✓ Manufacturing processes

    Complex tasks:

    - ✓ Fraud detection
    - ✓ Autonomous driving
    - ✓ Smart energy management
    - ✓ Industrial robotics

  - Benefits of AI Automation

    - ✓ Increased speed and productivity
    - ✓ Reduced human error
    - ✓ Lower operational costs
    - ✓ Continuous operation (24/7 systems)
    - ✓ Improved decision accuracy
    - ✓ Automation is a major reason AI is transforming industries.

b. Intelligent decision-making systems.

AI is not only about learning patterns — it is also about making decisions based on analyzed data. An **intelligent decision-making system** can:

- Evaluate information
- Compare alternatives
- Choose the best action
- Adapt to changing conditions

These systems are widely applied in engineering and technology.

- **Expert Systems and Recommendation Systems**

  - **Expert Systems**

    An expert system is an AI program designed to mimic the decision-making ability of a human expert. It uses:

    - ✓ Knowledge base (facts and rules)
    - ✓ Inference engine (reasoning process)

    Examples:

    - ✓ Medical diagnosis systems
    - ✓ Troubleshooting computer hardware
    - ✓ Engineering fault detection
    - ✓ Expert systems are common in industrial applications.

  - **Recommendation Systems**

    Recommendation systems suggest products or services based on user behavior. They use machine learning and predictive models.

    Examples:

    - ✓ Netflix movie recommendations
    - ✓ YouTube video suggestions
    - ✓ Online shopping recommendations

    These systems improve user experience and increase engagement.

- **Robotics and Autonomous Systems**
  - **Robotics**

    Robotics combines AI with mechanical systems to build machines that can perform physical tasks. AI enables robots to:

    ✓ Sense the environment
    ✓ Make decisions
    ✓ Perform actions intelligently

    Applications:

    ✓ Industrial robots in factories
    ✓ Surgical robots in medicine
    ✓ Warehouse automation
  - **Autonomous Systems**

    Autonomous systems operate without human control. Examples include:

    ✓ Self-driving cars
    ✓ Drones
    ✓ Autonomous delivery robots
    ✓ Smart military systems

    They rely on AI for:

    ✓ Navigation
    ✓ Object detection
    ✓ Real-time decision-making
- **Natural Language Processing (NLP) and Computer Vision**
  - **Natural Language Processing (NLP)**

    NLP allows machines to understand, interpret, and respond to human language. It is used in:

    ✓ Chatbots and virtual assistants
    ✓ Voice recognition systems
    ✓ Language translation
    ✓ Sentiment analysis

    Examples:

    ✓ Siri, Alexa
    ✓ Google Translate
    ✓ Customer support bots

- **Computer Vision**

  Computer vision enables machines to interpret and analyze visual data such as images and videos. Applications include:

  - Facial recognition
  - Surveillance systems
  - Medical imaging
  - Autonomous driving object detection

  Computer vision is essential for intelligent machines that interact with the real world.

c. AI Integration in Computer Engineering

In Computer Engineering, AI is integrated with both **hardware and software** to develop intelligent systems such as:

- Smart embedded devices
- AI-powered microcontrollers
- Edge computing systems
- IoT with machine learning
- Robotics and automation controllers

AI systems can run on:

- CPUs and GPUs
- Embedded processors
- FPGA and ASIC accelerators

This creates faster, smarter, and more efficient computing solutions.